

Inspironics AIoT + GenAI Platform

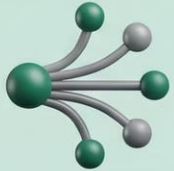
AWS Grant & Accelerator Submission

PREDICTIVE • COGNITIVE • SUSTAINABLE INFRASTRUCTURE



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Our Vision



Unified AIoT + GenAI Platform

one operating system for intelligent infrastructure.



**Transform facilities into autonomous,
self-optimizing systems.**



**Convergence of Energy, ESG,
and Experience.**

The Infrastructure Intelligence Gap



15–30% Energy Wasted — inefficiency across commercial and industrial facilities.



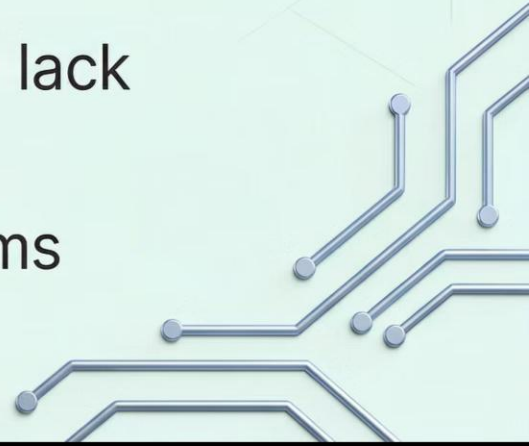
Reactive Maintenance — unplanned downtime erodes operational continuity.



No Cognitive Intelligence — systems lack predictive and adaptive capabilities.



Fragmented Control — siloed platforms with no unified management layer.



Three-Layer Intelligent Stack



Layer 1 — Device

Caleido Kombos (CK)

IoT edge devices and
real-time telemetry



Layer 2 — Intelligence

AIoT / Enervision

ML models, digital twins,
and energy analytics



Layer 3 — Cognition

Cielo Epic (CE)

GenAI cognitive layer for
language-driven insights

AWS Service Mapping

Platform Layer	AWS Services
 IoT	AWS IoT Core + Greengrass
 Data	Amazon S3 + Kinesis
 ML	Amazon SageMaker
 GenAI	Amazon Bedrock
 Application	Lambda + API Gateway

Track 1 — Caleido

Kombos: IoT + ML POC

Track Label



IoT sensor devices and telemetry ingestion



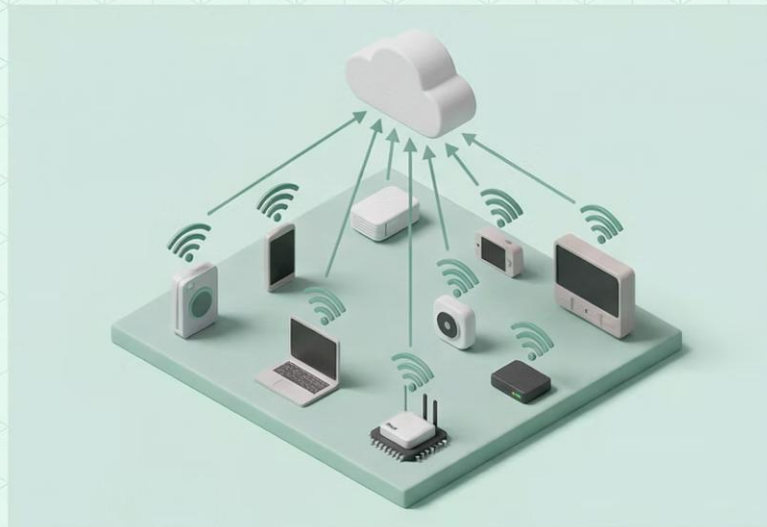
Real-time infrastructure monitoring



Edge-to-cloud architecture



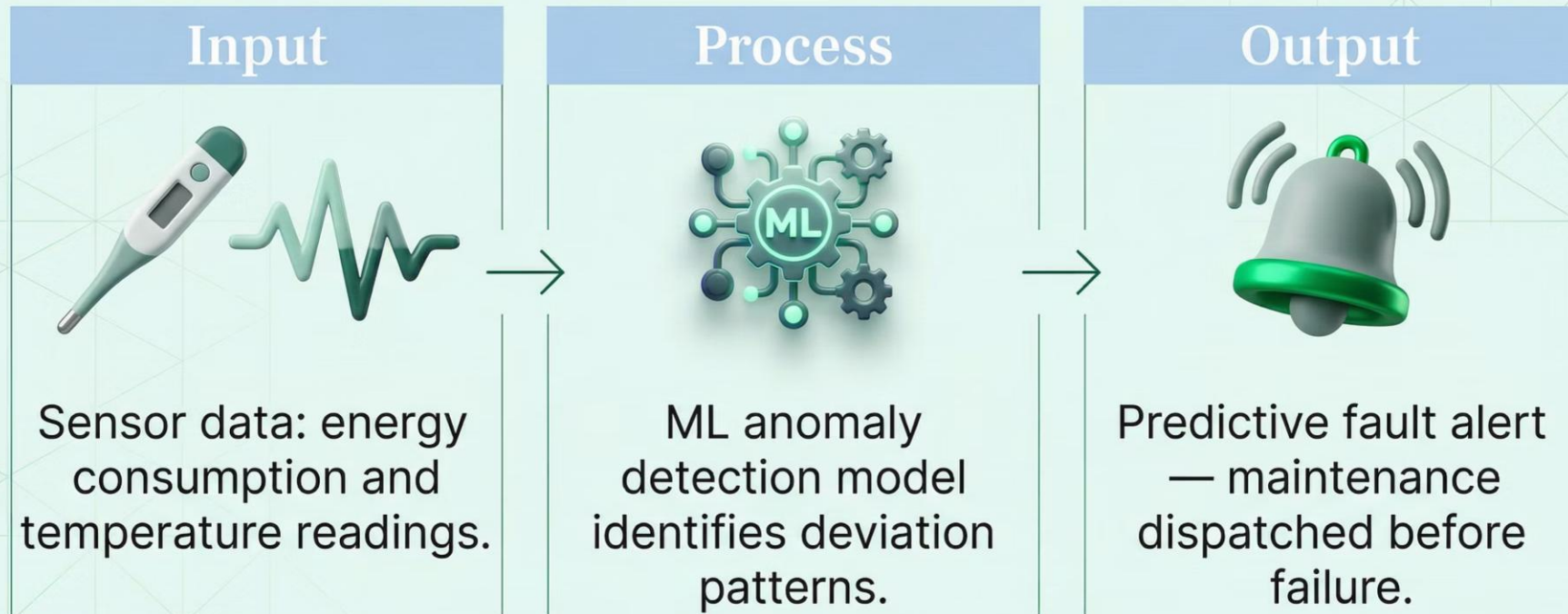
ML-ready data pipeline



Track 1 — End-to-End Data Flow



Track 1 Use Case — HVAC Anomaly Detection



Track 2 — Cielo Epic: GenAI Cognitive Platform

Track label



Generative AI cognitive layer for infrastructure



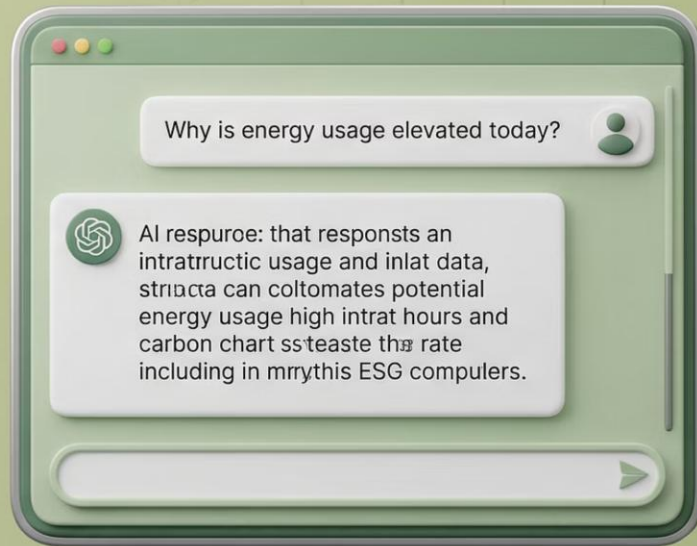
Natural language interaction with operational data



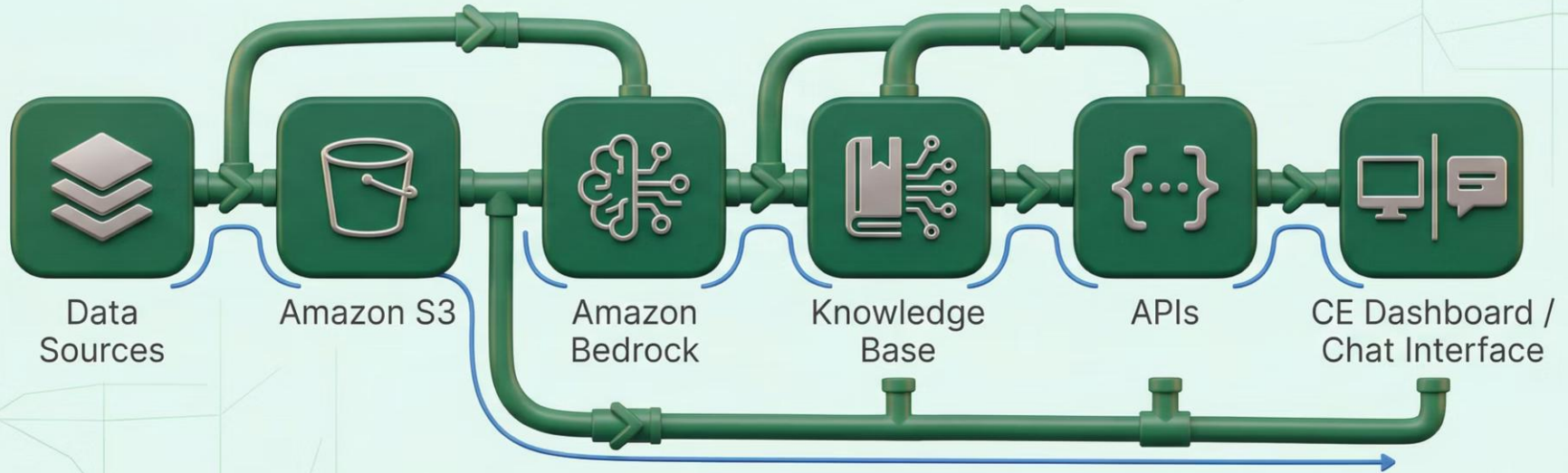
AI-driven decision intelligence



ESG insights and carbon reporting.



Track 2 — GenAI Data Flow



Cielo Epic — Conversational Intelligence in Action



Why is energy consumption high?

Root cause analysis via natural language.



Which devices may fail soon?

Predictive fault identification on demand.



ESG and Carbon Reporting

Automated sustainability metrics and compliance documentation.



Operational Recommendations

AI-suggested actions for efficiency and cost reduction.

Track 3 — AIoT / Enervision: Digital Twin Intelligence



Digital twin-powered real-time infrastructure modeling



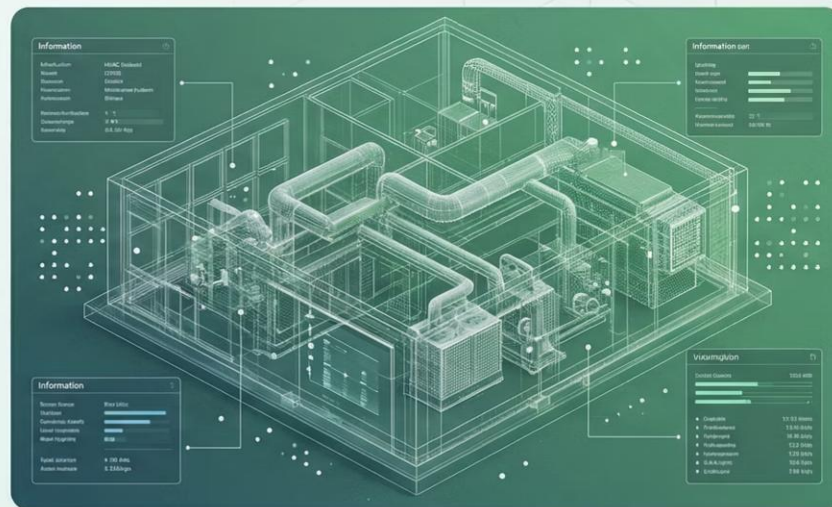
Predictive maintenance with ML foresight



Energy optimization across facilities



Deep infrastructure analytics and performance benchmarking



Quadrupole Intelligence Model

Q1 — Anomaly Detection

Identify irregular patterns in real time.



Q2 — Predictive Maintenance

Forecast failures before they occur.

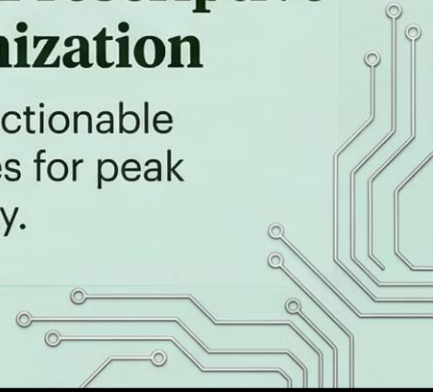
Q3 — Preventive Scheduling

Optimize maintenance timing and resource allocation.

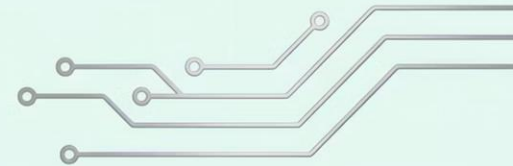
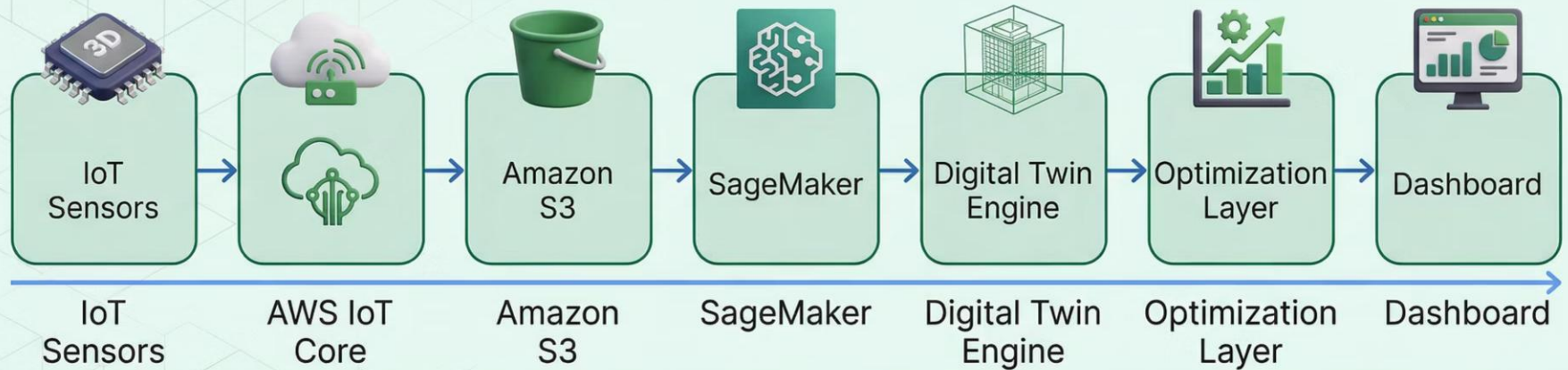


Q4 — Prescriptive Optimization

Deliver actionable strategies for peak efficiency.



Track 3 — Digital Twin Data Flow



Measured Outcomes — Built for Enterprise Scale



15–45%

Energy Cost Reduction

Validated across commercial facility types.



30–70%

Maintenance Cost Reduction

ML-driven predictive intervention.



< 12 Months

Projected Payback Period

Favorable unit economics at scale.



ESG Ready

Regulatory Compliance

Automated carbon and sustainability reporting.

Why We Build on AWS



End-to-End Stack — from IoT edge to GenAI cognitive layer in a single ecosystem.



Scalable IoT + ML — IoT Core and SageMaker purpose-built for industrial workloads.



GenAI via Bedrock — foundation model access with enterprise guardrails.



Global Infrastructure — deployment-ready across regions with marketplace reach.

Our Ask — AWS Partnership & Acceleration

What We Are Requesting

	AWS Credits — IoT Core, SageMaker, Bedrock, S3
	Technical Collaboration — co-engineering on ML and GenAI pipelines
	Go-to-Market Partnership — joint GTM with AWS enterprise channels
	Pilot Deployment Support — first facility deployment with AWS team

 Phase 1: POC
(0–3 mo)

 Phase 2: Pilot
(3–6 mo)

 Phase 3: Scale
(6–12 mo)

What Sets Inspironics Apart

Traditional BMS / SCADA

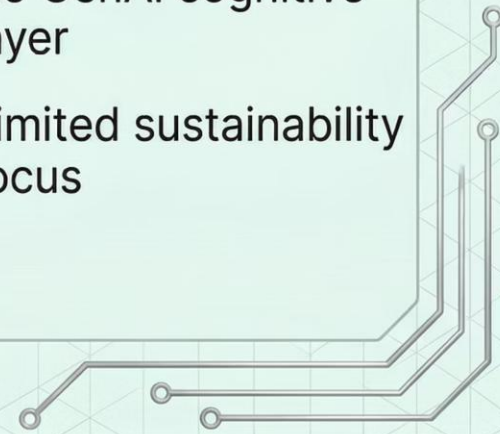
- Reactive only
- No AI layer
- Siloed data
- Manual reporting

Inspironics Platform

- ✓ Predictive + Prescriptive AI
- ✓ Unified IoT + GenAI stack
- ✓ Digital twin modeling
- ✓ Automated ESG reporting

Generic IoT Platforms

- No domain expertise
- No GenAI cognitive layer
- Limited sustainability focus



Leadership & Advisory Strength



Founder / CEO — AIoT & Infrastructure Strategy



CTO — ML Systems & Cloud Architecture



Head of Product — Enterprise SaaS & ESG.



Deep domain expertise in building systems and energy tech



Prior enterprise deployments across Asia-Pacific



AWS-aligned technical stack from inception.



Ecosystem Partners & Validation.

Intelligent Infrastructure. Measurable Impact.

PREDICTIVE • COGNITIVE • SUSTAINABLE

Inspironics — AWS Grant & Accelerator Submission

✉ contact@inspironics.com

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